AR Sign Language

AI-Powered Speech and Text to Sign Language Translation using 3D Avatar in Unity

PROJECT REPORT

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Abstract

ARsign Language is an AI-powered communication system developed to bridge the communication gap between hearing-impaired individuals and normal speakers. The system converts speech and typed text into sign language using a realistic 3D animated avatar in Unity.

The application includes a "Hold to Speak" feature for voice input and a text input system for manual typing. Speech is converted into text using AI-based speech recognition, and the processed text is translated into sign-language gestures using animation mapping.

Unlike traditional prerecorded sign-language video systems, ARsign Language uses real-time avatar animation for better scalability, interaction, and accessibility.

The project demonstrates the integration of Unity, AI, speech recognition, and immersive interaction technologies to create an innovative communication platform.

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1. Executive Summary

ARsign Language is a prototype AI-powered communication platform that translates speech and text into sign language using a 3D avatar. The application supports hearing-impaired individuals by combining speech recognition, text processing, Unity-based avatar animation, and AI-driven interaction. Users can provide input through a 'Hold to Speak' feature or by typing text. The processed text is converted into sign-language gestures performed by a 3D avatar in real time.

The application combines speech recognition, Unity avatar animation, text processing, and AI-driven interaction systems. Users can either speak using a “Hold to Speak” button or type text into the input field.

The processed text is converted into sign-language gestures performed by a 3D avatar in real time. The system demonstrates how AI and immersive technologies can improve communication accessibility.

2. Introduction

Communication barriers between hearing-impaired individuals and others remain a significant social challenge. Many people are unfamiliar with sign language, making communication difficult in schools, hospitals, workplaces, and public environments. Traditional solutions often rely on prerecorded videos or human interpreters, which can be difficult to scale. ARsign Language addresses this challenge by converting speech and text into sign-language animations using AI and Unity.

Traditional sign-language systems often rely on prerecorded videos or human interpreters. These systems are limited and difficult to scale.

ARsign Language addresses this problem using a dynamic avatar-based system. The application converts speech and text into sign-language animations using Unity and AI technologies.

3. Problem Statement

Hearing-impaired individuals often face communication challenges because many people do not understand sign language. Existing solutions can be expensive, non-interactive, or dependent on prerecorded content. This project aims to provide an intelligent system that converts spoken and typed language into sign language in real time using an interactive 3D avatar.

There is a strong need for an intelligent system capable of converting spoken and typed language into sign language in real time using a visual and interactive approach.

4. Objectives

- Convert speech into text using AI speech recognition.
- Process typed and spoken language.
- Translate text into sign-language gestures.
- Animate a 3D avatar using Unity.
- Build a real-time communication system.
- Improve accessibility using AI technologies.

5. Conceptualization

The project follows a human-centered, accessibility-focused design approach inspired by the communication challenges faced by deaf and hearing-impaired communities. It provides an intuitive platform where users can communicate naturally through speech or text.

The project focuses on providing an intuitive and interactive platform where users can communicate naturally using speech or text.

6. Literature Survey

Existing sign-language translation systems commonly use machine learning, computer vision, or prerecorded videos. Modern technologies such as Whisper, Unity, and avatar animation enable more interactive solutions. ARsign Language combines AI-based speech recognition with real-time avatar animation in a single prototype.

Modern technologies such as Whisper, Unity, MediaPipe, and avatar animation systems have enabled more interactive communication platforms.

ARsign Language combines AI speech recognition and real-time avatar animation into a single prototype system.

7. System Architecture

The system architecture is modular and scalable. Each module performs a specific task and passes its output to the next stage of processing.

Workflow:

Fig

Speech/Text Input → Speech Recognition → Text Processing → Sign Mapping → Unity Animation Controller → Avatar Output

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8. Technical Implementation

Unity is used to create the 3D environment and animate the avatar, while Python, Flask, and Whisper support speech recognition and backend processing.

The project uses:
• Unity Engine
• Python
• Whisper
• Flask API
• Mixamo Animations
• Ready Player Me Avatar

Speech input is processed using Whisper, and the resulting text triggers avatar animations in Unity.

9. Design & Assets

The application uses a semi-realistic avatar to improve gesture visibility and user engagement. The interface is designed to be simple, intuitive, and accessible.

Assets used:
• Ready Player Me for avatar creation
• Mixamo for animations
• Unity UI Toolkit for interface design

The UI was designed with simplicity and accessibility in mind.

10. User Interaction

Users can interact with the system using either the 'Hold to Speak' button or a text input field. After processing the input, the avatar performs the corresponding sign-language gestures in sequence.
1. Hold to Speak button
2. Text Input field

After processing the input, the avatar performs corresponding sign-language gestures in sequence.

11. Testing Plan

The project includes:
• Functional Testing
• User Testing
• Performance Testing

Testing verifies accurate speech recognition, smooth avatar animations, responsive UI behaviour, and reliable interaction flow.

12. Challenges Faced

1. Speech Recognition Noise:
Background noise affected speech accuracy.

2. Animation Synchronization:
Smooth transitions between avatar gestures were difficult.

3. Text-to-Sign Mapping:
Sentence structures differed from sign-language structures.

13. Team Contributions

The team collaborated on UI development, Unity integration, speech recognition, avatar animation, and testing. The project strengthened our understanding of AI integration, accessibility-focused design, and immersive technologies.

The project helped the team learn AI integration, accessibility-focused design, and immersive interaction systems.

14. Future Enhancements

Future versions may include:
• Finger-level animation tracking
• Advanced NLP
• Multiplayer communication
• Mobile AR support
• Larger datasets
• Cloud integration

15. Final Results

The final prototype successfully converts speech and text into sign-language animations using a 3D avatar, demonstrating the practical integration of AI, Unity, and accessibility technologies.

The project demonstrates the practical integration of AI, Unity, and accessibility technologies.

16. Conclusion

ARsign Language demonstrates how AI-powered technologies can improve communication accessibility for hearing-impaired individuals. The project integrates speech recognition, NLP, Unity-based animation, and avatar interaction into a real-time communication platform.

The project successfully integrates speech recognition, Unity animation, and avatar interaction into a real-time communication platform.

17. References

Unity Official Documentation

OpenAI Whisper Documentation

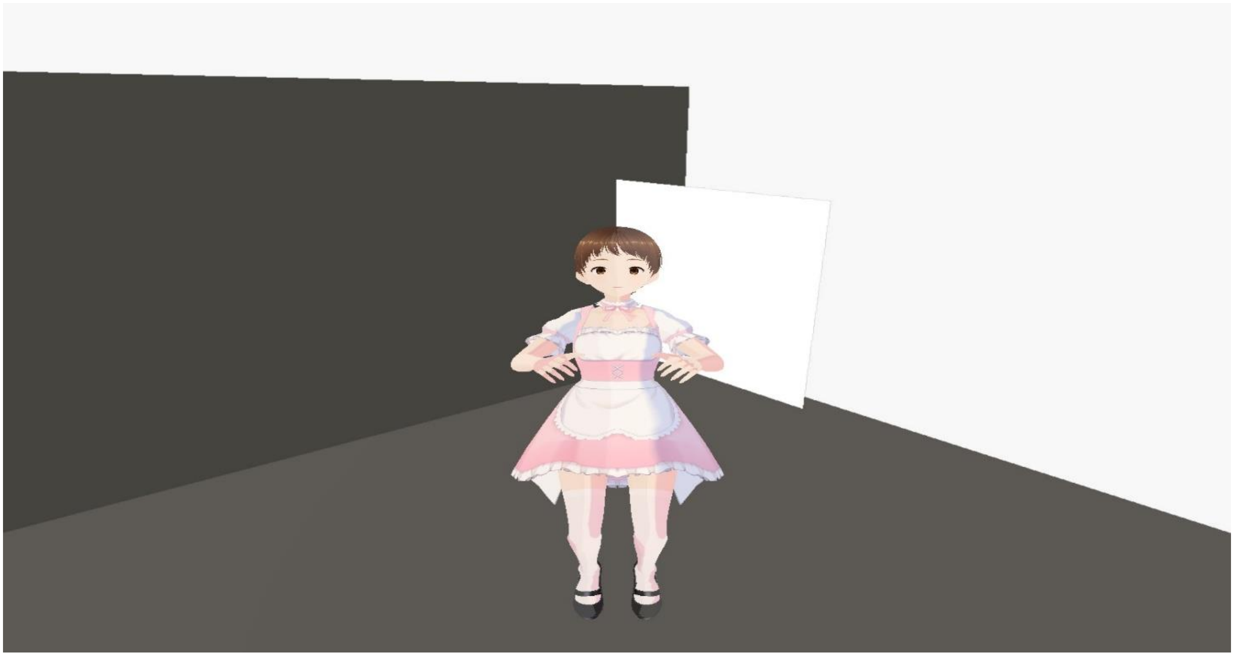
Flask Documentation

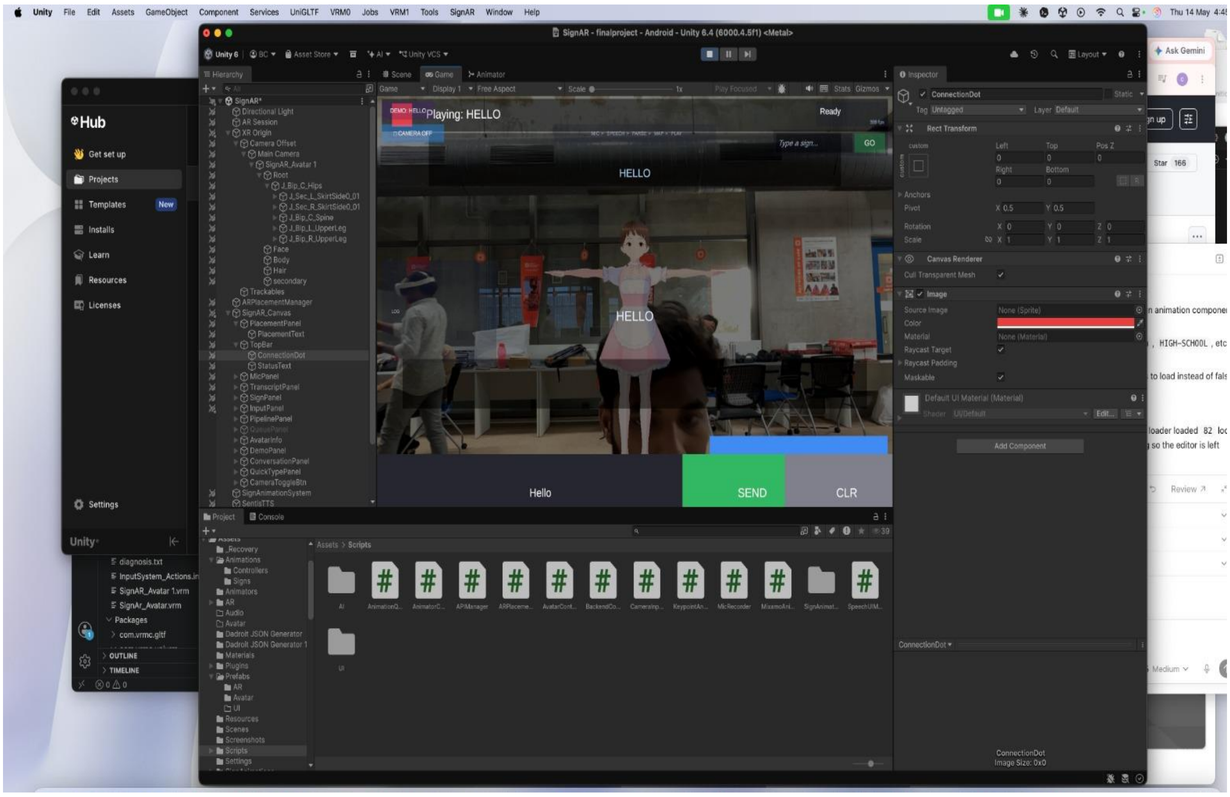
Mixamo Animation Library

Ready Player Me Avatar Platform

MediaPipe Documentation

18. Links of Project





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